Real-time vehicle Identification (SilentTrack)

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Introduction & Background

SilentTrack is an Automated License Plate Recognition (ALPR) system designed for covert police vehicles. Unlike traditional ALPR systems that capture every passing vehicle, SilentTrack selectively identifies only vehicles with previous reports (e.g., stolen, under investigation). This reduce data overload and supports real-time, discreet decision making during undercover operations.

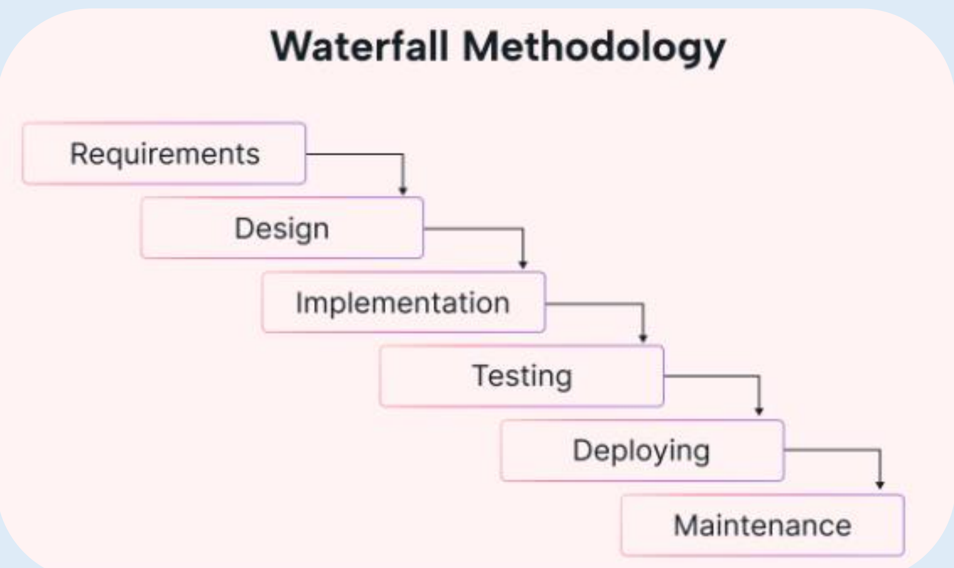
Traditional ALPR systems overwhelm officers with irrelevant data and often rely on cloud servers, which is unsuitable for mobile or covert missions, SilentTrack leverages high-resolution cameras, real-time OCR, embedded processing, and a secure local database to perform targeted vehicle recognition with high accuracy, even in low- light or fast motion scenarios.

Objective

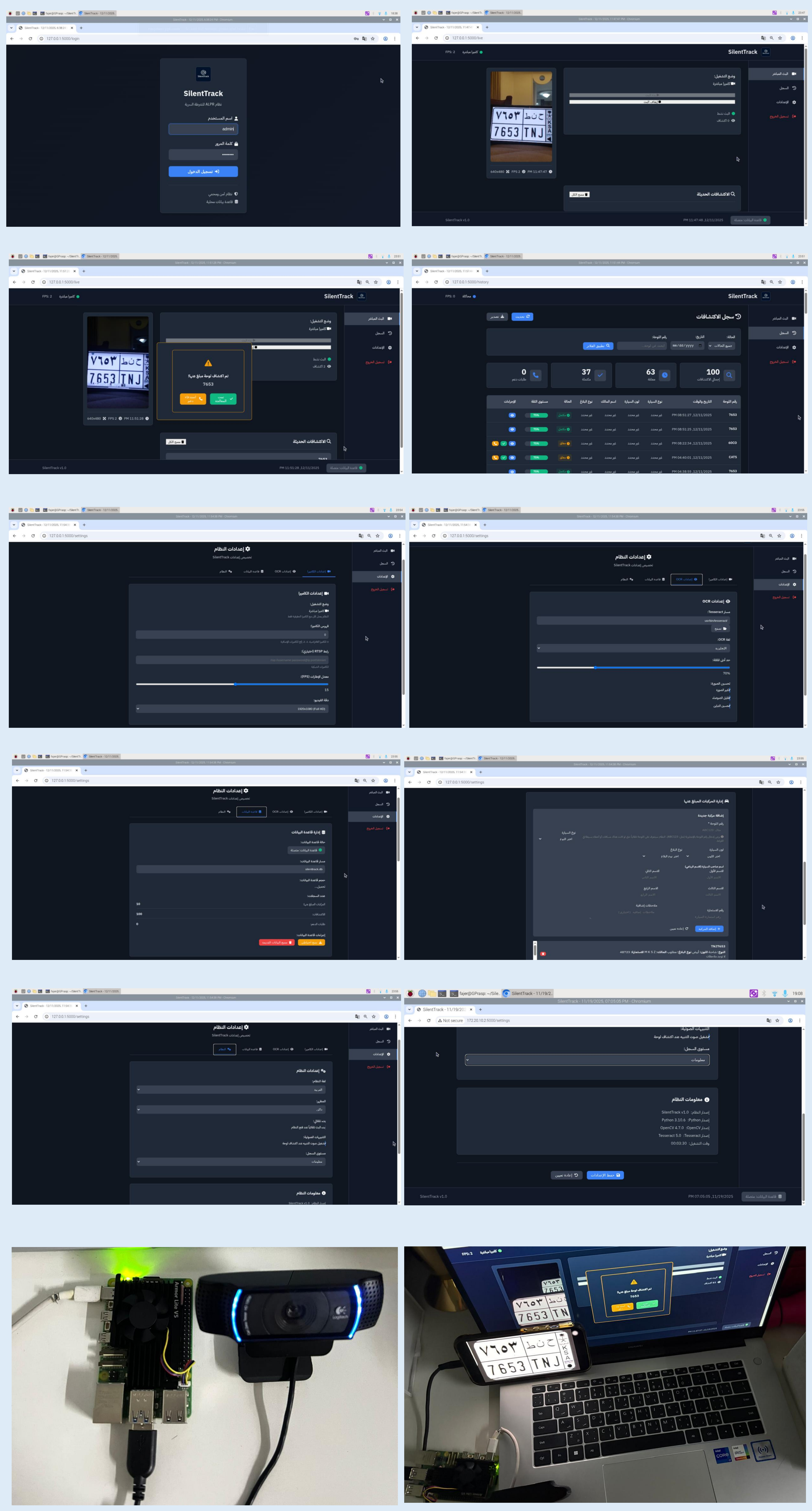
- Develop a selective, real-time ALBR system for secret police patrol.
- Reduce irrelevant data by focusing only on reported vehicles.
- Improve response time and operational efficiency.
- Ensure online capability with secure local storge.
- Achieve reliable OCR accuracy under different environmental condition.

Methodology

The system was developed using a structured and sequential process that began with understanding the needs of covert police operations and identifying the limitations of existing ALPR systems. After defining the system requirements, the architecture was designed to combine real-time image processing, OCR, and a secure local database. The implementation focused on integrating the camera feed with the processing unit to detect and recognize license plates instantly. Once the system was built, it was tested in different lighting and motion conditions to ensure fast and reliable performance suitable for field use



Interfaces



Tools

Raspberry Pi

python

OpenCV

Tesseract OCR

SQLite

logitech

conclusion

SilentTrack proved to be an effective solution for covert vehicle monitoring by providing quick, selective, and secure recognition of important vehicles. It improved response time and reduced information overload, making it suitable for real-world police operations. The system lays a strong foundation for future enhancements such as deeper AI integration and advanced situational awareness features.

Result

The system achieved fast and accurate license plate recognition, even in low-light and high-motion environments. It successfully identified reported vehicles in real time and displayed alerts clearly through the interface. Testing showed that the system reduced unnecessary data and maintained a focused, efficient workflow for officers during patrol.

Acknowledgment

First and foremost, we extend our deepest gratitude to Almighty ALLAH, whose blessings, guidance, and mercy have illuminated our path and empowered us to bring this project to completion. Without His will, none of this would have been possible.

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References